EXPERIMENT 8 MULTIRATE SIGNAL PROCESSING

8.1 Implementation and Analysis of Decimation and Interpolation in Multirate Signal Processing Using MATLAB

CODE

```
clc;
clear;
close all;
% Original Sequence
n = 0:40;
x = sin(0.3*pi*n);
% Decimation by 2
y = x(1:2:end);
nd = 0:length(y)-1;
% Interpolation by 2
v = zeros(1,2*length(x));
v(1:2:end) = x;
ni = 0:length(v)-1;
% Figure 1: Original Sequence
figure;
stem(n,x,'filled');
grid on;
title('Original Sequence x[n]');
xlabel('n');
ylabel('Amplitude');

% Figure 2: Decimated Sequence
figure;
stem(nd,y,'filled');
grid on;
title('Decimated Sequence by 2');
```

```
xlabel('n');
ylabel('Amplitude');

% Figure 3: Interpolated Sequence

figure;
stem(ni,v,'filled');
grid on;
title('Interpolated Sequence by 2');
xlabel('n');
ylabel('Amplitude');
% Figure 4: Comparison
figure;
stem(n,x,'b'); hold on;
stem(0:2:length(v)-2,v(1:2:end),'r');
legend('Original x[n]','Interpolated Samples');
grid on;
title('Comparison of Sequences');
xlabel('n');
ylabel('Amplitude');
```

OUTPUT

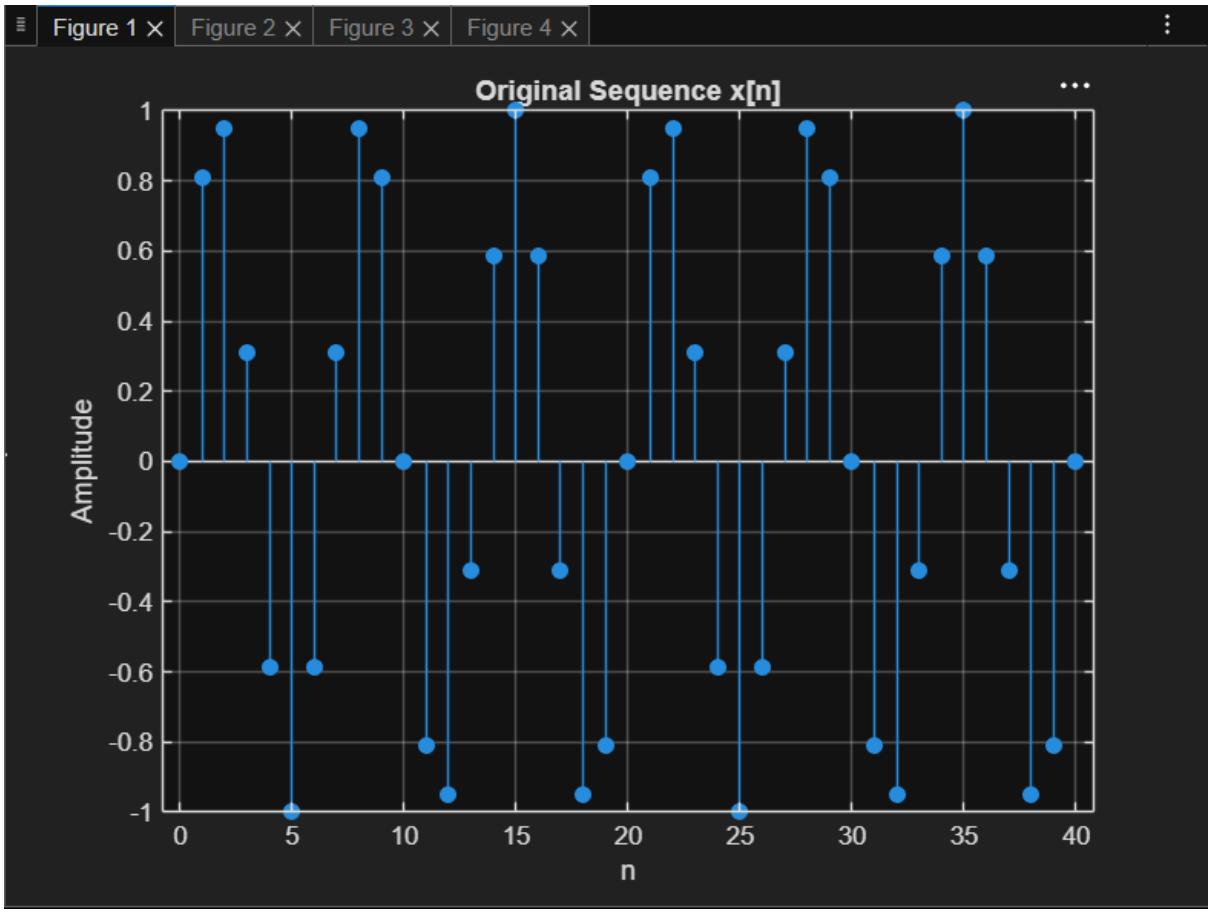


Figure 1
Figure 2
Figure 3
Figure 4
Original Sequence x[n]
Amplitude
n
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
0
5
10
15
20
25
30
35
40

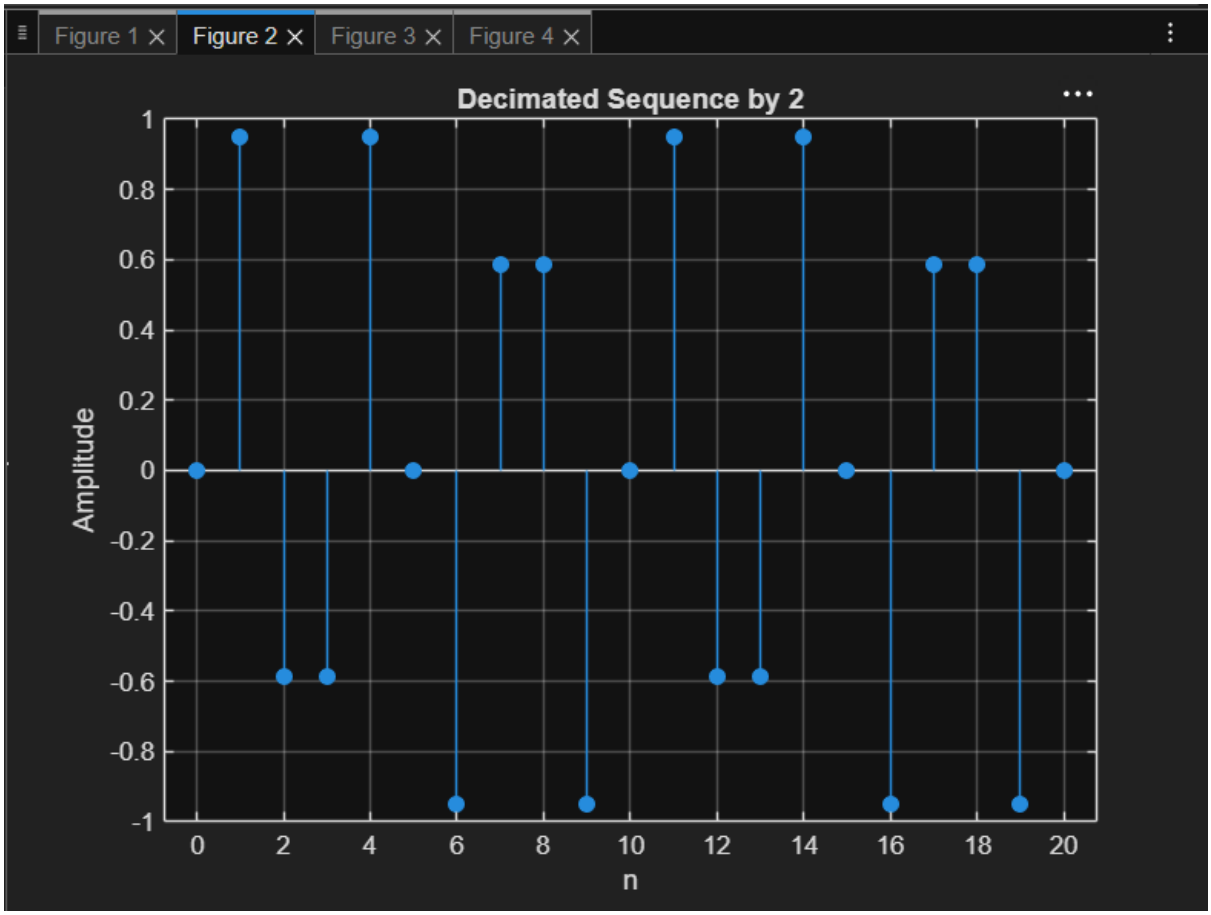


Figure 1
Figure 2
Figure 3
Figure 4
Decimated Sequence by 2
Amplitude
n
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
0
2
4
6
8
10
12
14
16
18
20

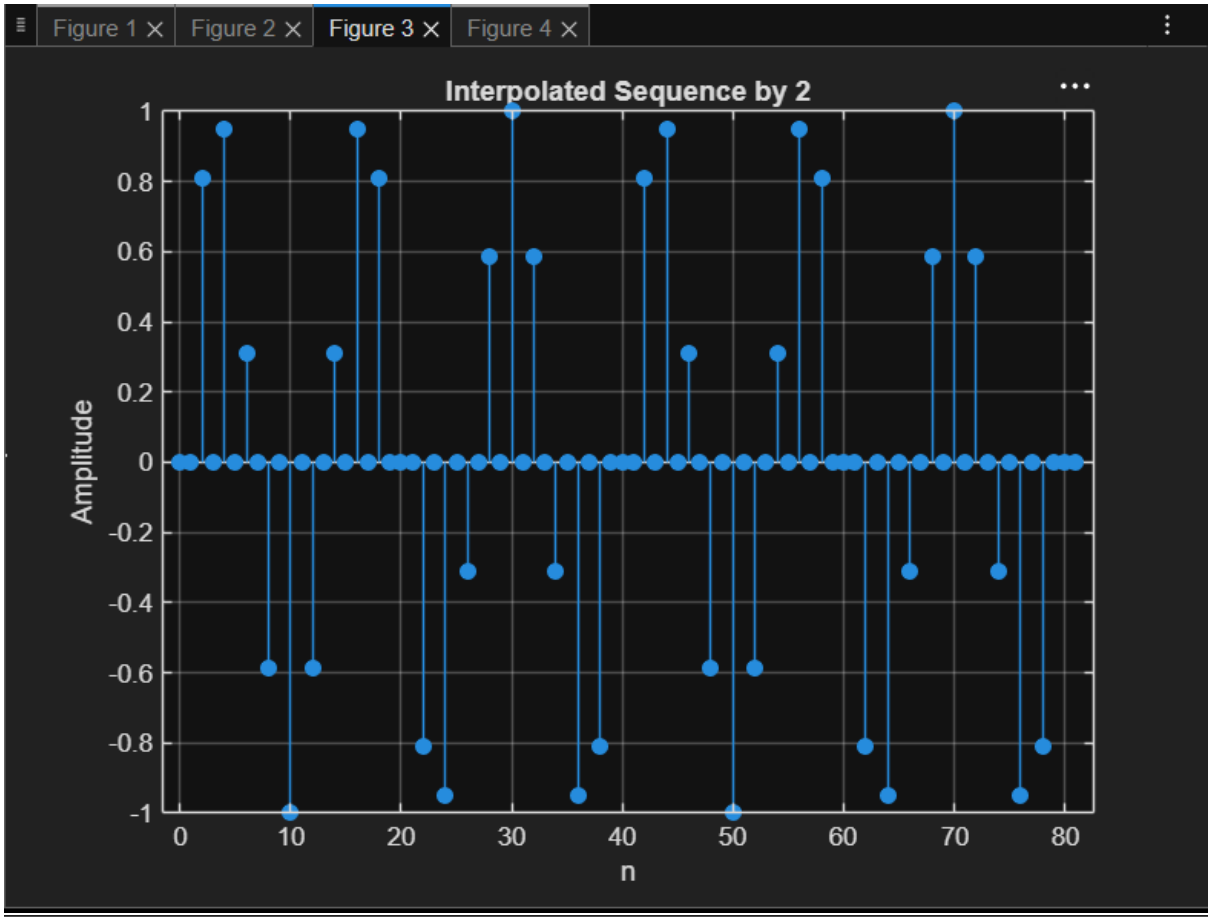


Figure 1
Figure 2
Figure 3
Figure 4
Interpolated Sequence by 2
Amplitude
n
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
0
10
20
30
40
50
60
70
80

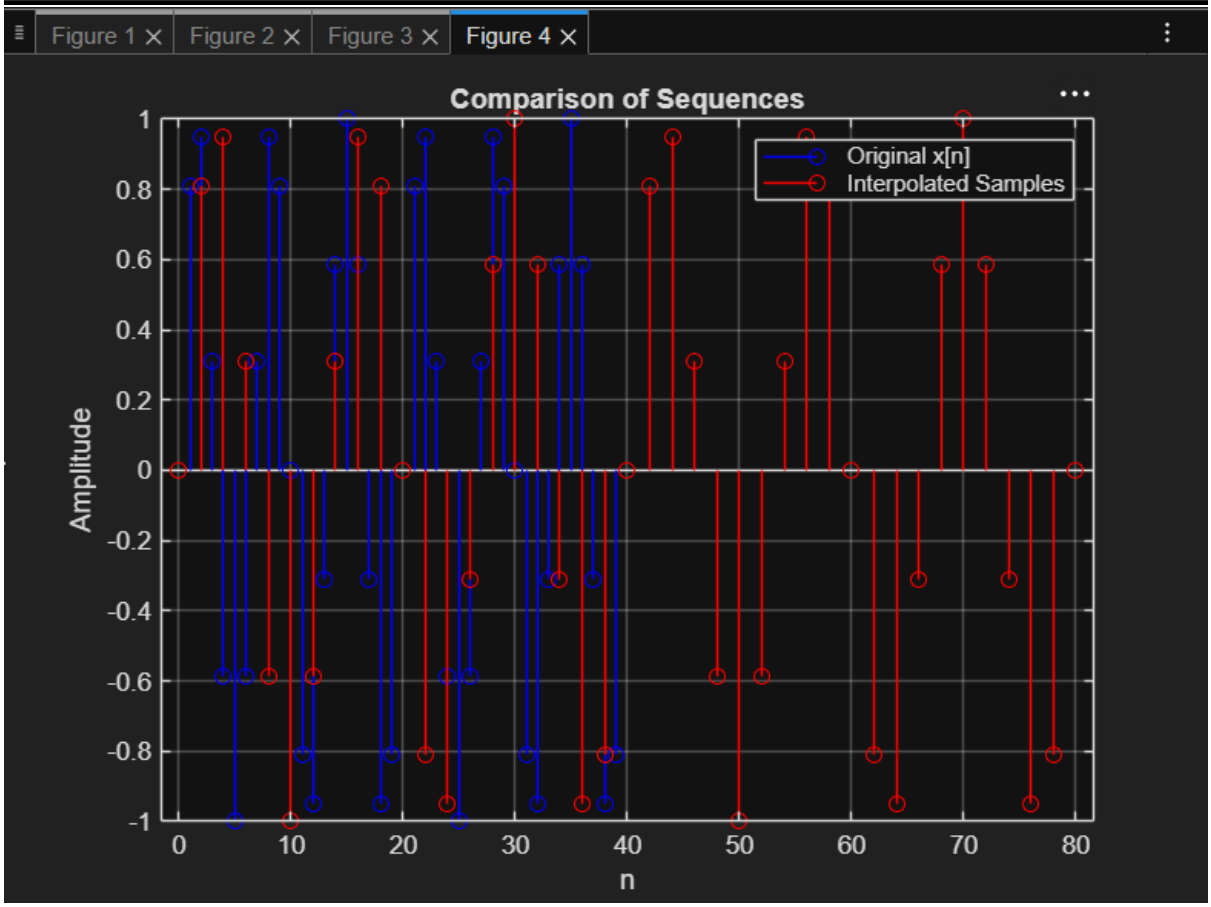


Figure 1
Figure 2
Figure 3
Figure 4
Comparison of Sequences
Original x[n]
Interpolated Samples
Amplitude
n
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
0
10
20
30
40
50
60
70
80

8.2 Polyphase Implementation of Decimation and Interpolation Filters Using MATLAB

CODE

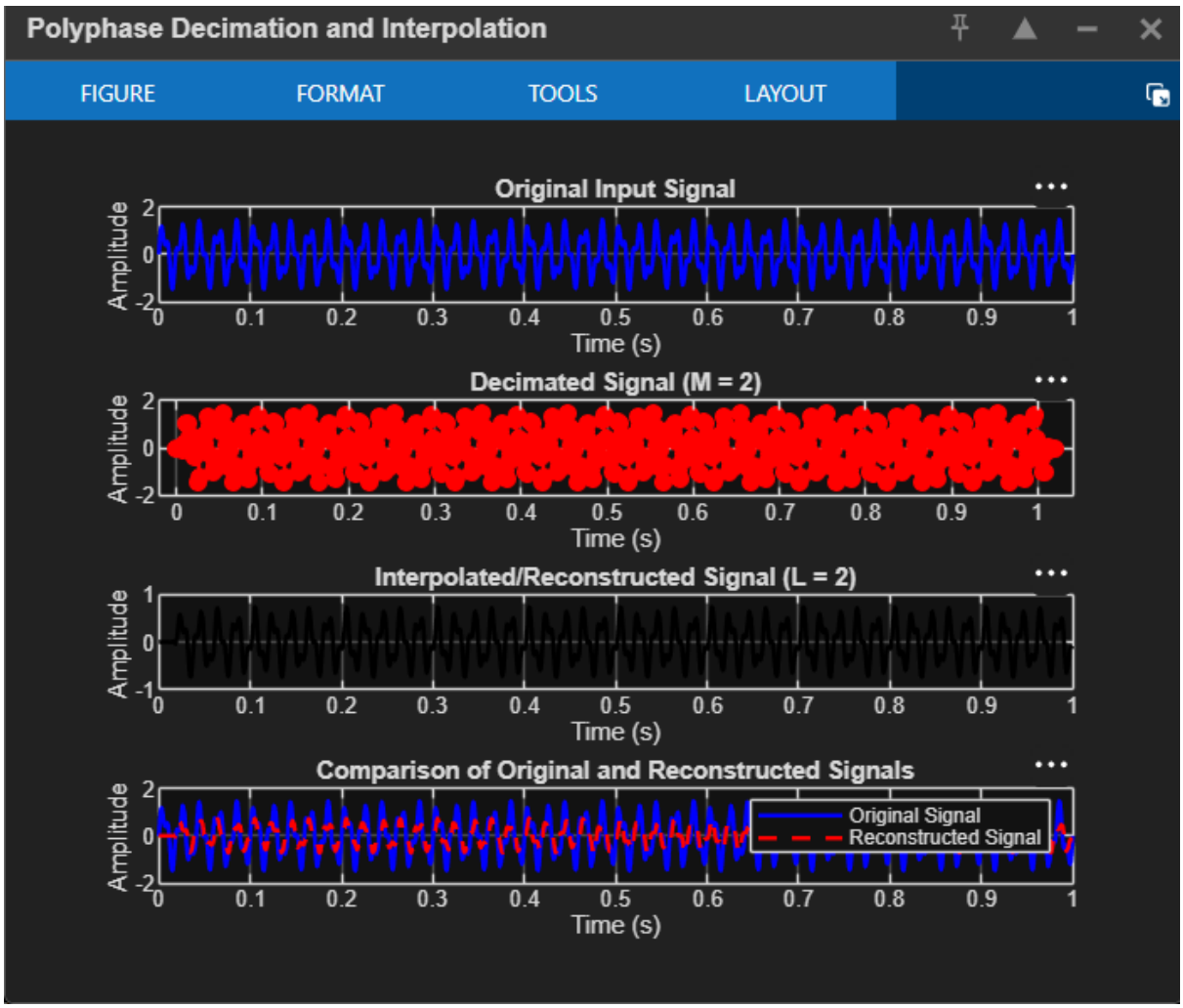
```
clc;
clear;
close all;
%% Input Signal
Fs = 1000; % Sampling Frequency
t = 0:1/Fs:1; % Time Vector
% Test Signal
x = sin(2*pi*50*t) + 0.5*sin(2*pi*120*t);

%% FIR Low-Pass Filter Design
N = 20; % Filter Order
Fc = 0.4; % Normalized Cutoff Frequency
h = fir1(N,Fc,hamming(N+1));
%% Decimation Factor
M = 2;
% Polyphase Decimation
y_dec = upfirdn(x,h,1,M);
%% Interpolation Factor
L = 2;
% Polyphase Interpolation
y_int = upfirdn(y_dec,h,L,1);
%% Adjust Length for Comparison
if length(y_int) > length(x)
    y_int = y_int(1:length(x));
elseif length(y_int) < length(x)
```

```
    x = x(1:length(y_int));
end
%% Time Axes
t1 = (0:length(x)-1)/Fs;
t2 = (0:length(y_dec)-1)/(Fs/M);
t3 = (0:length(y_int)-1)/Fs;
%% BEST GRAPH OUTPUT
figure('Name','Polyphase Decimation and Interpolation','NumberTitle','off');
% Original Signal
subplot(4,1,1)
plot(t1,x,'b','LineWidth',1.5)
title('Original Input Signal')
xlabel('Time (s)')

ylabel('Amplitude')
grid on
% Decimated Signal
subplot(4,1,2)
stem(t2,y_dec,'r','filled')
title('Decimated Signal (M = 2)')
xlabel('Time (s)')
ylabel('Amplitude')
grid on
% Interpolated Signal
subplot(4,1,3)
plot(t3,y_int,'k','LineWidth',1.5)
title('Interpolated/Reconstructed Signal (L = 2)')
```

```
xlabel('Time (s)')
ylabel('Amplitude')
grid on
% Comparison
subplot(4,1,4)
plot(t1,x,'b','LineWidth',1.5)
hold on
plot(t3,y_int,'r--','LineWidth',1.5)
title('Comparison of Original and Reconstructed Signals')
xlabel('Time (s)')
ylabel('Amplitude')
legend('Original Signal','Reconstructed Signal')
grid on
%% Display Information
disp('--------------------------------------');
disp('POLYPHASE DECIMATION & INTERPOLATION');

disp('--------------------------------------');
fprintf('Original Signal Length = %d\n',length(x));
fprintf('Decimated Signal Length = %d\n',length(y_dec));
fprintf('Interpolated Signal Length = %d\n',length(y_int));
%% Separate Figure for Better Visualization
figure;
plot(t1,x,'b','LineWidth',2)
hold on
plot(t3,y_int,'r--','LineWidth',2)
title('Original vs Reconstructed Signal')
```

```
xlabel('Time (s)')
ylabel('Amplitude')
legend('Original Signal','Reconstructed Signal')
grid on
```

OUTPUT



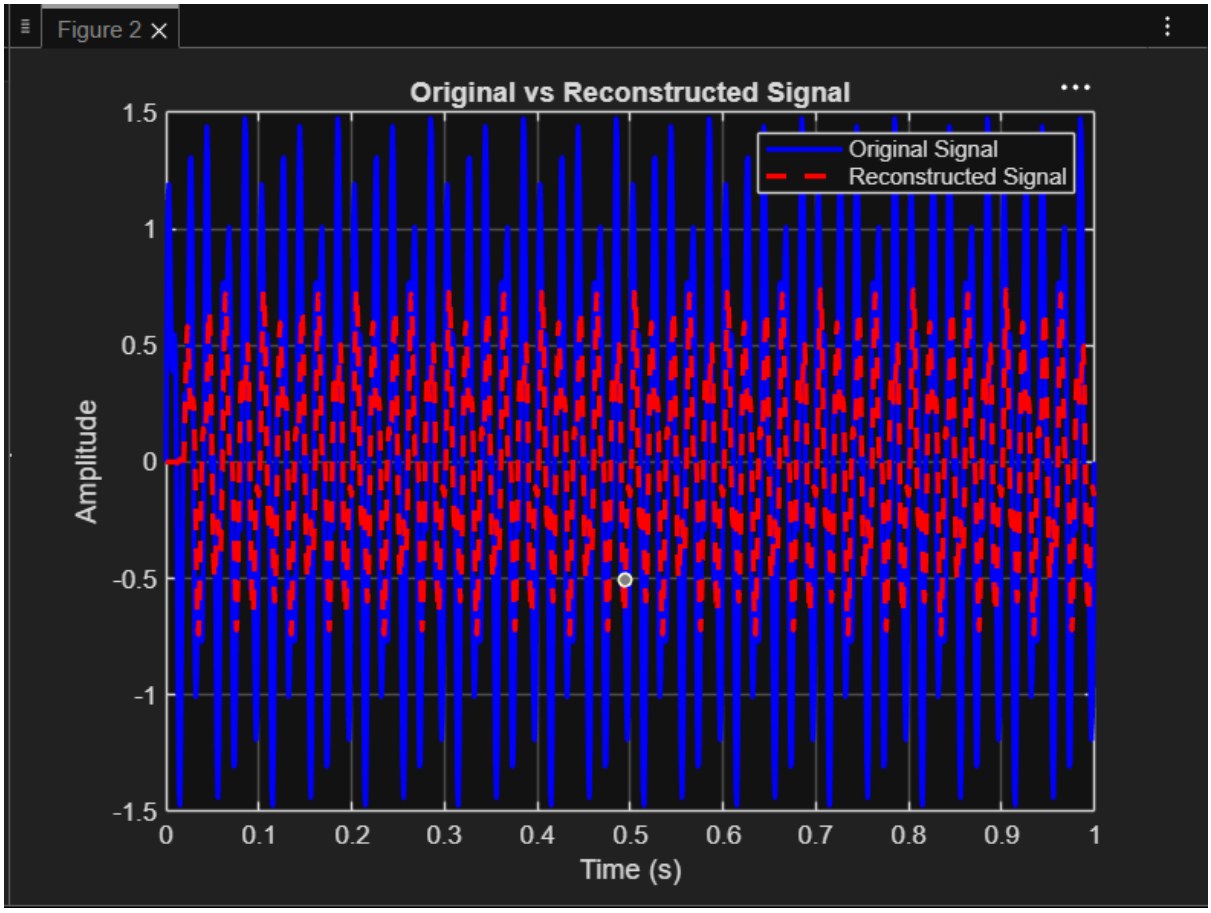


Figure 2
Original vs Reconstructed Signal
Original Signal
Reconstructed Signal
Amplitude
Time (s)
1.5
1
0.5
0
-0.5
-1
-1.5
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1